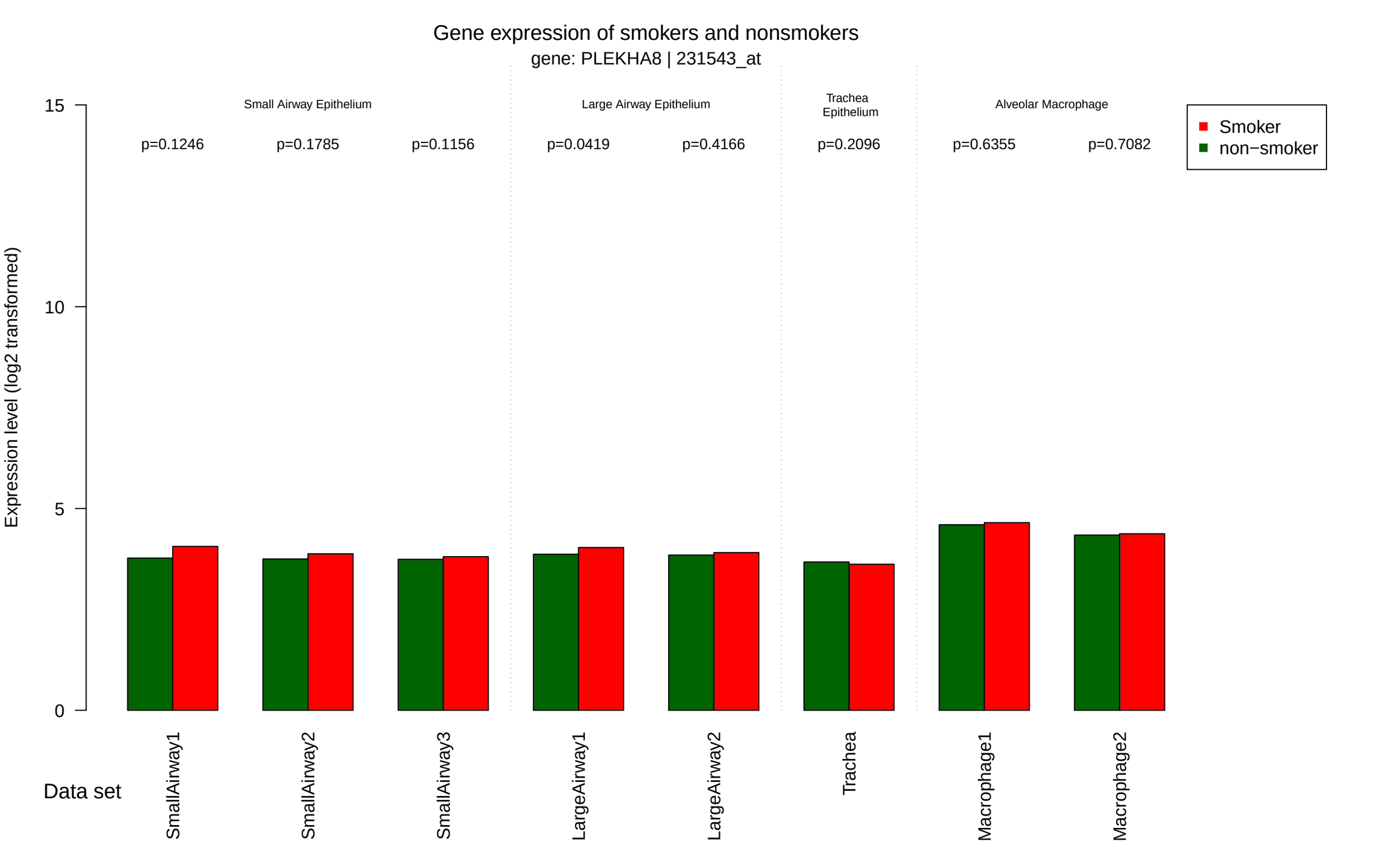
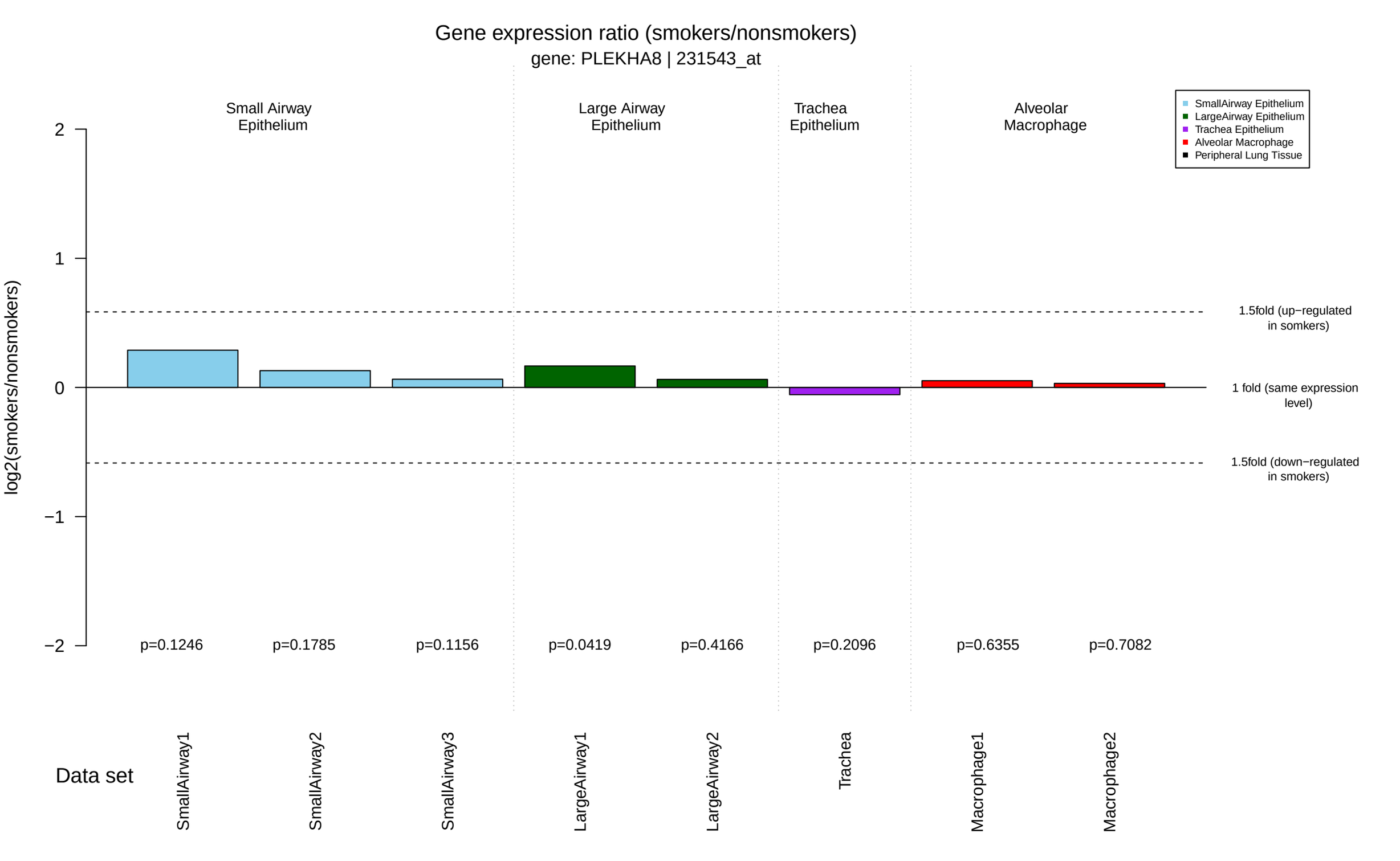
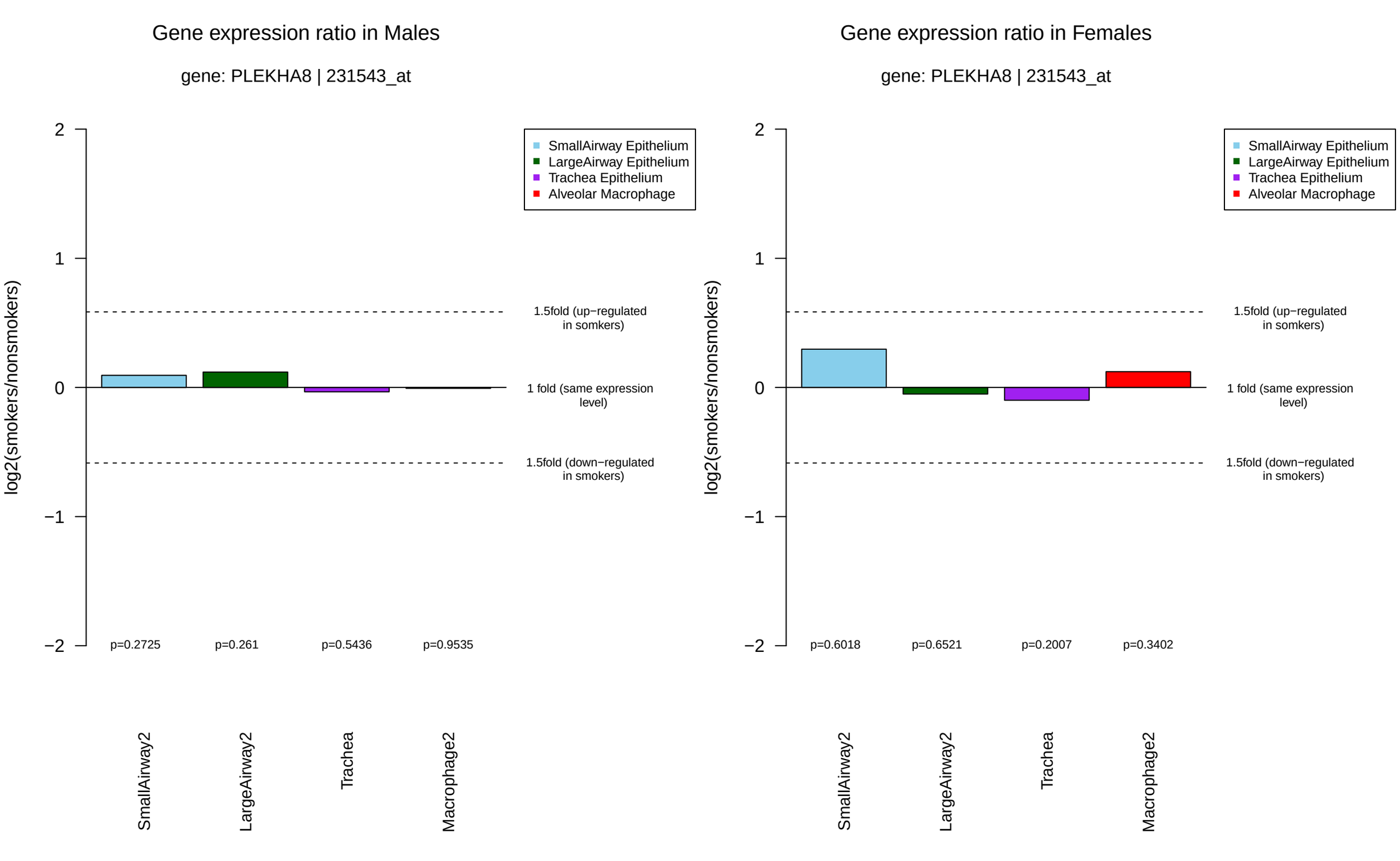
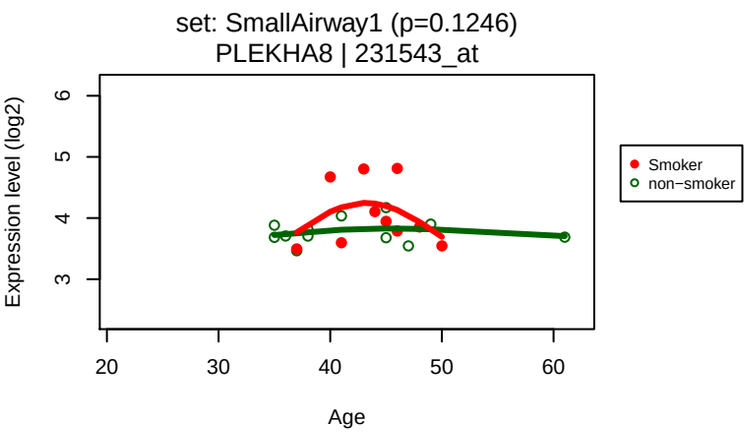
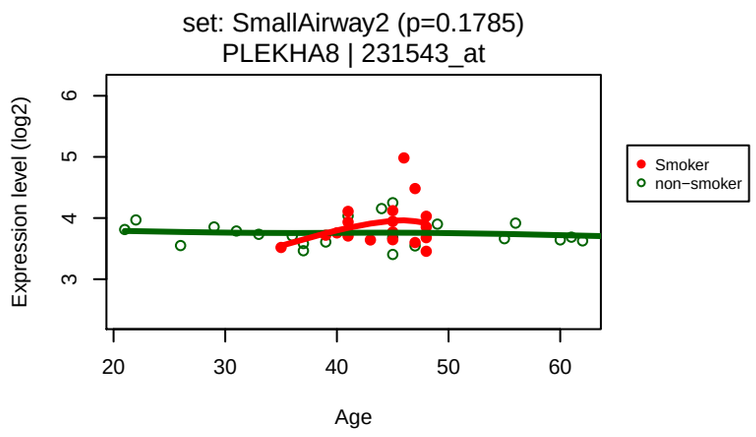
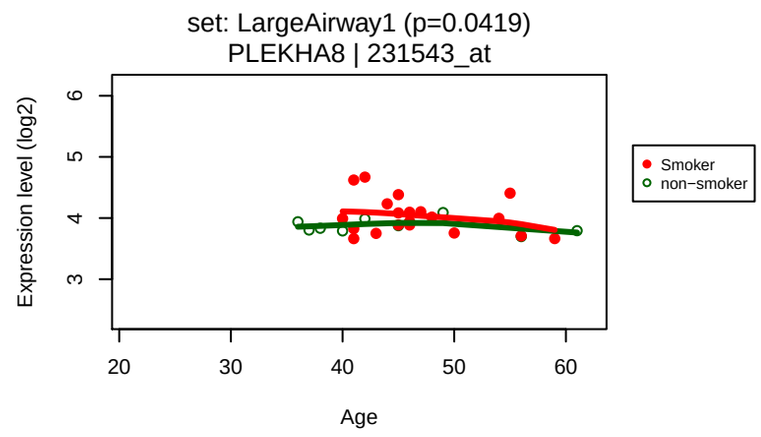

Gene expression of smokers and nonsmokers
gene: PLEKHA8 | 231543_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1246
p=0.1785
p=0.1156
p=0.0419
p=0.4166
p=0.2096
p=0.6355
p=0.7082
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

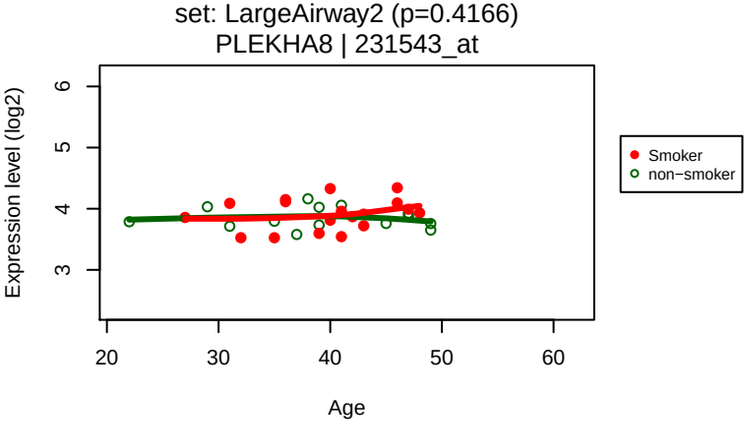

Gene expression ratio (smokers/nonsmokers)
gene: PLEKHA8 | 231543_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1246
p=0.1785
p=0.1156
p=0.0419
p=0.4166
p=0.2096
p=0.6355
p=0.7082
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

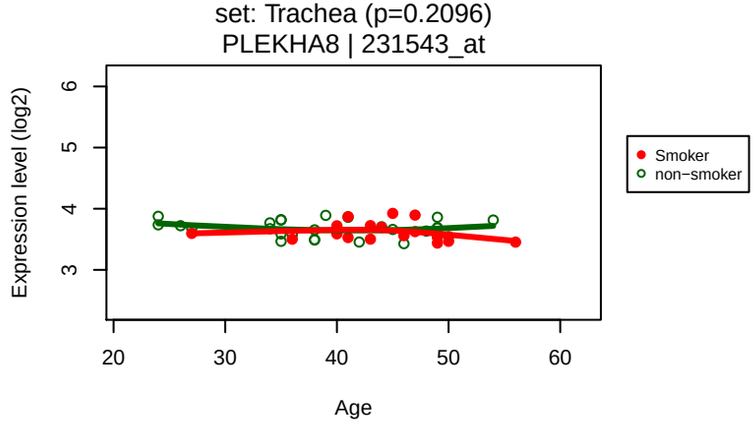

Gene expression ratio in Males
gene: PLEKHA8 | 231543_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2725
p=0.261
p=0.5436
p=0.9535
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PLEKHA8 | 231543_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6018
p=0.6521
p=0.2007
p=0.3402
SmallAirway2
LargeAirway2
Trachea
Macrophage2

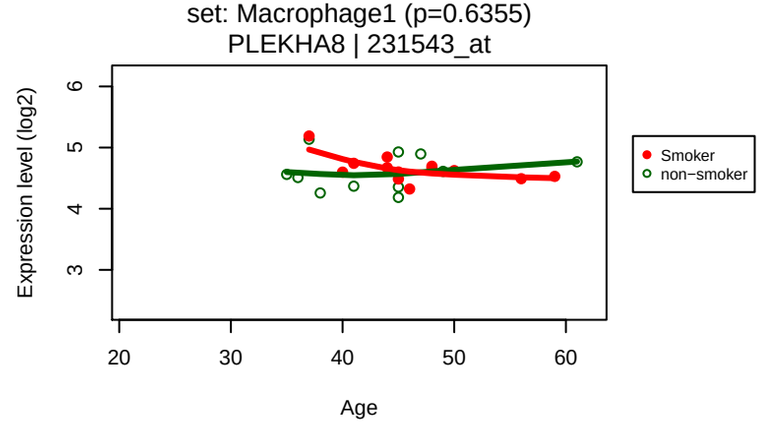

set: SmallAirway1 (p=0.1246)
PLEKHA8 | 231543_at
Expression level (log2)
Age
Smoker
non−smoker

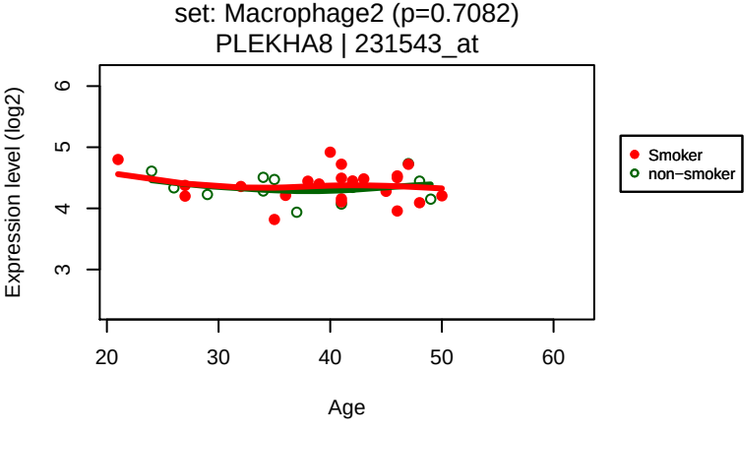

set: SmallAirway2 (p=0.1785)
PLEKHA8 | 231543_at
Expression level (log2)
Age
Smoker
non−smoker

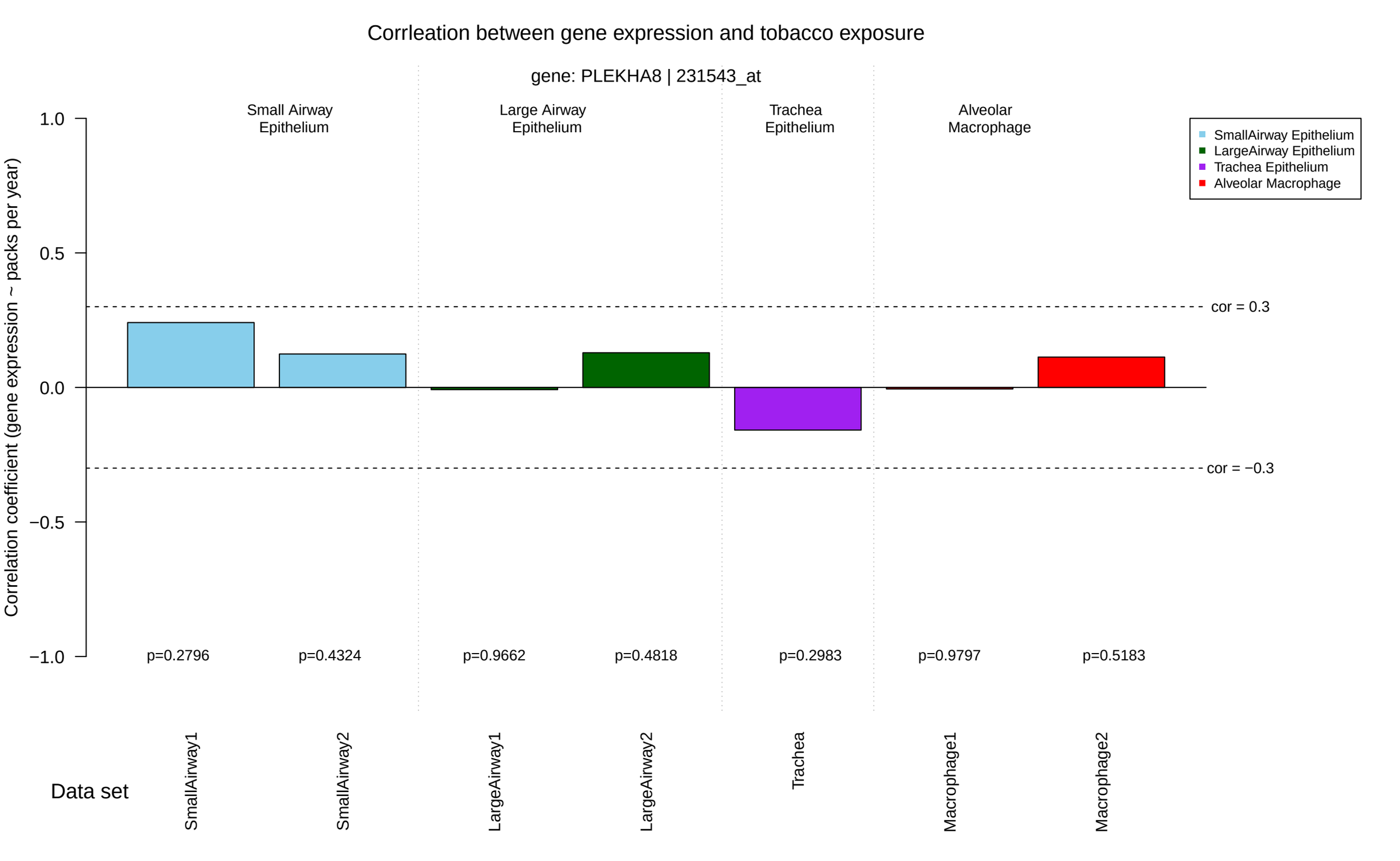

set: LargeAirway1 (p=0.0419)
PLEKHA8 | 231543_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4166)
PLEKHA8 | 231543_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2096)
PLEKHA8 | 231543_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6355)
PLEKHA8 | 231543_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7082)
PLEKHA8 | 231543_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PLEKHA8 | 231543_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2796
p=0.4324
p=0.9662
p=0.4818
p=0.2983
p=0.9797
p=0.5183
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2